Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics

Experience

Data Scientist — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production applications and data pipelines for client businesses via Upwork and Fractal, specializing in predictive analytics, automated ETL ingestion, and CI/CD testing pipelines.
- **Order Procurement & Management Solution** | *Python, PySpark, PostgreSQL, SQL, TypeScript*
  - Built a full-stack order management portal utilizing Python, SQL, PostgreSQL, and TypeScript with session-based authentication and RESTful data APIs.
  - Designed a dual-tier order interface with a customer portal for real-time inventory listings and an admin suite for task dispatch, audit logs, and recurring order analytics.
  - Engineered an automated data ingestion ETL pipeline using Python and PySpark to parse unstructured WhatsApp messages into relational database records, accelerating order throughput.
  - Optimized relational PostgreSQL schemas with indexing and SQL query constraints to handle 1,000+ daily order transactions and analytics efficiently.

Computer Vision Engineer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *Python, OpenCV, C++, Django, Flutter* **(Live link)**
  - Architected a native C/C++ bridge via dart:ffi to port OpenCV Scale-Invariant Feature Transform (SIFT) algorithms into Flutter for cross-platform computer vision across Android and iOS.
  - Constructed a real-time camera feed pipeline that normalized OS frame buffers and aspect ratios, converting raw video streams into C structs with zero memory leaks and 30+ FPS performance.
  - Implemented real-time keypoint feature extraction and object classification, transmitting match coordinates asynchronously to Dart for spatial augmented reality UI overlay rendering.
  - Created responsive mobile interfaces in Flutter from Figma designs and built RESTful backend APIs in Django with Python to power location-aware spatial analytics features for alpha release.
- **Kia National Conference App** | *Flutter, Python, SQL*
  - Deployed an event management mobile app for 500+ attendees at the Kia National Dealer Conference in Bangkok, featuring real-time registration, SQL-backed attendee directories, and venue navigation.

Technical Skills

Languages & Data Science: Python, SQL, R, PySpark, C++, C, JavaScript/TypeScript, Mathematica
Machine Learning & Analytics: Machine Learning, OpenCV (SIFT), Scikit-learn, Pandas, NumPy, Seaborn
Generative AI & Frameworks: Generative AI and LLMs, RAG, Prompt Engineering, Django, PostgreSQL, React
MLOps, Cloud & Tools: MLOps and CI/CD, AWS, Docker, Git, Linux, Bash, LaTeX, Agentic Development

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)